Exercises for Image Processing

1. Explain different type of images in terms of colour depth, such as 1-bit images, 8-bit gray-level images and 24-bit colour images.

2. Describe the process of face detection from an image. Explain why the Haar cascade detector can be effective and computationally efficient for this problem.

3. Name three different methods for image segmentation, and briefly describe how they work.

4. What is the difference between semantic segmentation and instance segmentation?

5. Discuss the traditional feature-based object detection methods and the modern neural network based (in particular deep learning based) object detection methods.